PRKDC (DNA-PKcs) — Phylogenetic Distribution & the Efferocytosis Macrophage-Proliferation Route

Gene: PRKDC / DNA-PKcs (human) · **UniProt:** P78527 **Family:** PANTHER **PTHR11139:SF68** ("DNA-dependent protein kinase catalytic subunit"); OrthoDB `8724497at2759` (Eukaryota), `5996771at33208` (Metazoa) **Focus:** free_text hypothesis `prkdc-animal-distribution` · **Taxon:** *Homo sapiens* (NCBITaxon:9606) **Investigation:** 3 iterations · 4 confirmed findings · 2 primary papers

Summary

Verdict: Partially supported — the specific loss claim is verified, but the framing over-reaches, and the result is curation-irrelevant for human PRKDC.

The seed hypothesis bundles an evolutionary claim with a mechanistic premise. The mechanistic premise — that the efferocytosis-induced macrophage-proliferation route involves DNA-PKcs sensing apoptotic-cell DNA — is **supported** by the primary source (Gerlach et al. 2021, PMID: 34784501), with one refinement: the activating ligand is **DNase2a-derived nucleotides**, not intact double-strand DNA ends, and DNA-PKcs acts upstream of an **mTORC2/Rictor→Myc** axis in resolving macrophages. The evolutionary claim — that DNA-PKcs is absent from *Drosophila melanogaster* and *Caenorhabditis elegans* — is **directly verified**. However, the implied narrative that absence characterizes "some animal lineages" broadly is **overstated**: DNA-PKcs is an **ancient, metazoan-ancestral gene** retained across nearly all animal clades, with genuine losses confined narrowly to *Drosophila*, *Caenorhabditis*, and (outside animals) budding yeast.

Across three iterations, two independent orthology resources (OrthoDB and a per-organism UniProt/PANTHER PTHR11139 family census) converged on the same picture. The OrthoDB orthogroup places DNA-PKcs in 287 of 5,768 sampled eukaryotes, overwhelmingly single-copy, including non-vertebrate metazoans. The UniProt/PANTHER census then confirmed that fly and worm — both complete, gold-standard genomes — retain the sister kinases of the PIKK

family (TRRAP, mTOR, ATR, SMG1) but have specifically lost the DNA-PKcs subfamily (SF68), while the mosquito *Anopheles* (same insect order as *Drosophila*) retains it. A phylum-broad census further showed SF68 present from sponges and cnidarians through most insects, and revealed that earlier OrthoDB insect "absences" (beetle, bee, silkmoth) were **sampling artifacts, not losses**.

Bottom line for the curator: this is a genuine and interesting evolutionary observation, but it provides **no basis to add, remove, generalize, or specialize any GO term on human PRKDC**. Ortholog presence/absence in flies and worms is a property of those genomes, not evidence about the human gene product's molecular function, biological process, or cellular component. The item should be logged as **non-core background/evolutionary context**, and the efferocytosis route (Gerlach 2021) treated as a context-specific, downstream role rather than a core human function.

Key Findings

Finding 1 — DNA-PKcs is an ancient eukaryotic gene lost independently in specific lineages, not vertebrate-restricted

The historical assumption that DNA-PKcs is a vertebrate-specific PIKK arose largely because it is absent from the two most-studied invertebrate models. The orthology data refute this reading. The OrthoDB orthogroup matching PANTHER PTHR11139:SF68 and UniProt P78527 (Eukaryota node `8724497at2759`; Metazoa node `5996771at33208`) has a phyletic profile of **present in 287 of 5,768 sampled eukaryotic species**, and is **overwhelmingly single-copy (283/287)**. Among the metazoan carriers are non-vertebrate representatives such as the cephalochordate *Branchiostoma lanceolatum*, the echinoderm *Acanthaster planci*, and the dipteran insects *Anopheles merus* and *Bradysia odoriphaga*. *Drosophila melanogaster*, *Caenorhabditis elegans*, and *Saccharomyces cerevisiae* are confirmed **absent**.

This broad-but-patchy, single-copy pattern is the hallmark of an ancient gene subject to independent secondary losses, not of a late vertebrate innovation. It is directly corroborated by Lees-Miller et al. 2021 (PMID: 33035590), who report that DNA-PKcs "is widely distributed in invertebrates, fungi, plants, and protists." The fly/worm absences are therefore best read as lineage-specific losses against an ancient background — not as evidence of a narrow, derived distribution.

Finding 2 — The efferocytosis route uses DNA-PKcs to sense DNase2a-derived nucleotides (context-specific, not a new GO assignment)

The mechanistic premise is accurate but requires precision about the stimulus. Gerlach et al. 2021 (PMID: 34784501) showed that, after engulfment of apoptotic cells, **nucleotides derived from the hydrolysis of apoptotic-cell DNA by phagolysosomal DNase2a activate a DNA-PKcs–mTORC2/Rictor pathway that increases Myc to promote non-inflammatory macrophage proliferation**. Silencing macrophage DNase2a or Rictor blocked efferocyte proliferation in both a dexamethasone-thymus model and an atherosclerosis-regression model.

Two nuances matter for curation. First, the proximal stimulus is **degraded DNA/nucleotides**, not intact double-strand DNA ends — mechanistically distinct from DNA-PKcs's canonical role binding double-strand breaks during non-homologous end joining (NHEJ). Second, the role is **cell-type- and context-specific** (resolving/efferocytic macrophages), demonstrated in mouse and in vitro. It is a downstream signaling function layered on a specialized cell state, not a constitutive core activity of the gene product.

Finding 3 — Drosophila and C. elegans absence is a true subfamily-specific loss; mosquito retains it

To exclude the possibility that fly/worm absences were database gaps, an independent census of the ATM-related parent family **PTHR11139** was run per organism via UniProt+PANTHER. Human carries the DNA-PKcs subfamily **PTHR11139:SF68** (P78527, 4,128 aa) alongside mTOR (SF9), ATR (SF69), SMG1 (SF71), and TRRAP (SF1). *Drosophila melanogaster* has **8 PTHR11139 members but zero SF68** (present: TRRAP/SF1, mTor/SF9, Smg1/SF119, ATR/SF69). *C. elegans* has **5 PTHR11139 members but zero SF68** (TRRAP, smg-1, ATR, TOR). Because both genomes are complete and the sister kinases are present, SF68 absence is a **genuine gene loss, not an unsequenced/unannotated genome**.

Critically, *Anopheles gambiae* retains one SF68 member (Q7Q293, 4,191 aa). The DNA-PKcs loss is therefore **lineage-specific within Diptera** — *Drosophila* lost it while mosquitoes kept it — rather than a general feature of flies or insects.

Finding 4 — DNA-PKcs is ancestral to Metazoa and retained by most insects; verified losses are narrowly restricted

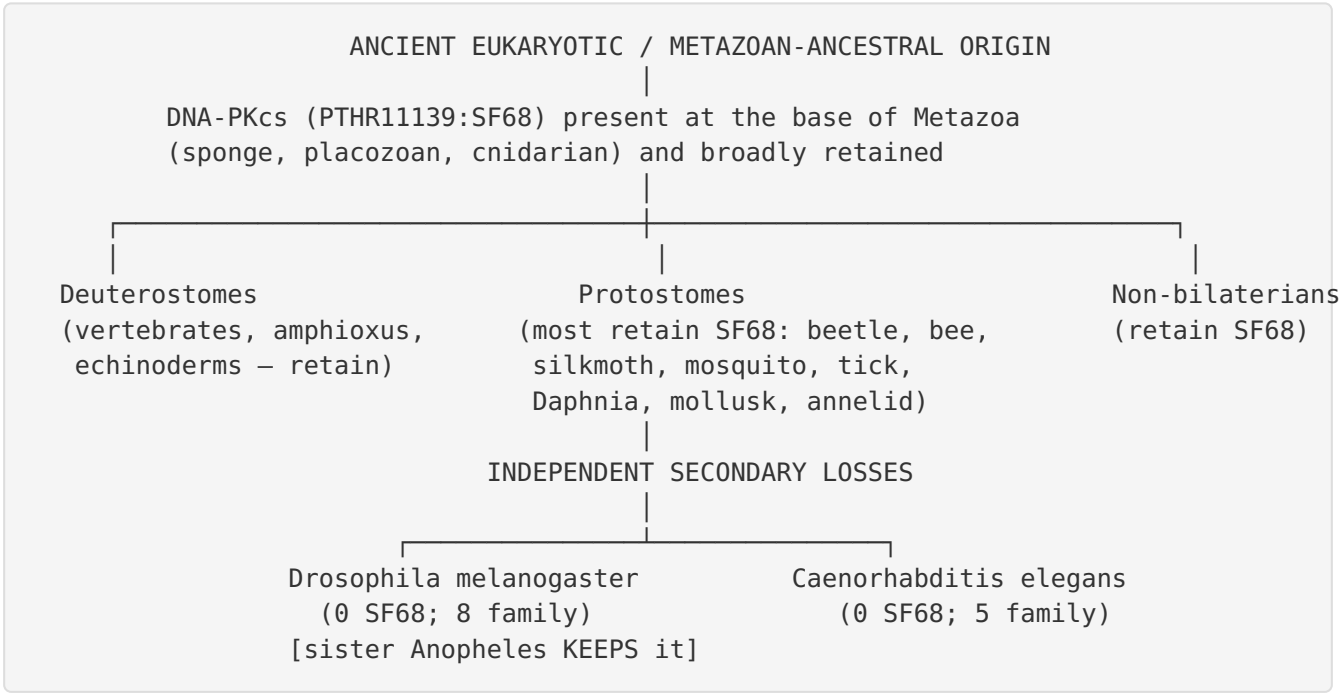
A phylum-broad UniProt+PANTHER PTHR11139 census across 14 additional metazoan proteomes places the origin of DNA-PKcs at the base of the animal tree. SF68 is **present** in the sponge *Amphimedon queenslandica*, the placozoan *Trichoplax adhaerens*, the cnidarians

Nematostella vectensis and *Hydra vulgaris*, the mollusk *Crassostrea gigas*, the annelid *Capitella teleta*, the echinoderm *Strongylocentrotus purpuratus*, the chelicerate tick *Ixodes scapularis*, the crustacean *Daphnia pulex*, and the insects *Tribolium castaneum* (beetle), *Apis mellifera* (bee), *Bombyx mori* (silkmoth), and *Aedes aegypti* (mosquito). Several of these insects had been **absent from the OrthoDB orthogroup**, confirming those absences were sampling artifacts rather than losses. *Octopus bimaculoides* returned zero PTHR11139 members total (incomplete UniProt proteome), so it is uninformative and must not be scored as a loss.

Taken together, the genuine, verified losses among animals are confined to **Drosophila melanogaster** and **Caenorhabditis elegans** (plus *Saccharomyces cerevisiae* outside the animals) — precisely the historical model organisms that seeded the now-refuted "vertebrate-specific" myth.

Mechanistic Model / Interpretation

The evidence separates cleanly into two threads the seed bundles together: an **evolutionary** thread and a **mechanistic/cell-biological** thread. Keeping them apart clarifies the curation implication.



Evolutionary thread. DNA-PKcs is old, single-copy, and pan-metazoan. The losses in *Drosophila* and *C. elegans* are real but narrow — two convergent, lineage-specific deletions coinciding with the two most popular invertebrate models, which created the false impression of vertebrate specificity. The mosquito/fly contrast within Diptera is the single cleanest demonstration that these are lineage-specific events, not clade-wide absences.

Mechanistic thread. In mammals, within efferocytic macrophages, DNA-PKcs senses not intact DNA ends but **DNase2a-derived nucleotides** produced by phagolysosomal degradation of ingested apoptotic-cell DNA, signaling through mTORC2/Rictor to raise Myc and drive non-inflammatory proliferation that resolves tissue injury. This is a specialized, context-dependent signaling role distinct from the canonical NHEJ/DNA-repair function that anchors human PRKDC's core GO annotations.

The seed's logical step — "no DNA-PKcs ⇒ route cannot operate" — is valid for *Drosophila* and *C. elegans*. But as a general statement about "some animal lineages," it is misleadingly broad: the route's molecular hardware is present almost everywhere in animals.

Thread	Claim	Status	Impact on human PRKDC GO
Evolutionary	Absent in fly and worm	**Confirmed (true loss)**	None
Evolutionary	Absent from "some animal lineages" broadly	**Overstated** — losses are narrow	None
Evolutionary	Vertebrate-specific	**Refuted** — ancient, pan-metazoan	None
Mechanistic	Senses apoptotic-cell DNA in efferocytosis	**Confirmed, with precision** (DNase2a-derived nucleotides)	Context-specific; non-core

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
PMID: 34784501 (Gerlach 2021)	Direct assay + mutant phenotype	Supports (with refinement)	DNA-PKcs mediates efferocytosis-induced proliferation via apoptotic DNA	DNase2a nucleotides → DNA-PKcs→mTORC2/Rictor→Myc; silencing DNase2a or Rictor blocks proliferation	Mouse resolving macrophages; thymus & atherosclerosis-regression models; in vitro
PMID: 33035590 (Lees-Miller 2021)	Structural/ evolutionary (phylogenomics)	Qualifies/ Refutes framing	DNA-PKcs vertebrate-restricted?	"Widely distributed in invertebrates, fungi, plants, and protists"	Pan-eukaryotic survey
OrthoDB `8724497at2759` / `5996771at33208`	Computational (orthology)	Supports losses / Qualifies framing	Metazoan distribution of SF68	Present 287/5768 eukaryotes, 283 single-copy; carriers incl. *Branchiostoma*, *Acanthaster*, *Anopheles merus*; absent fly/worm/yeast	OrthoDB current release
UniProt+PANTHER PTHR11139 census (fly/worm/mosquito/yeast)	Computational (2nd resource)	Supports losses; resolves sampling caveat	Fly/worm absence real?	Fly (8) & worm (5) carry TRRAP/mTOR/ATR/SMG1 but 0 SF68 (complete genomes → true loss); *Anopheles* retains SF68 (Q7Q293)	UniProt reference entries
UniProt+PANTHER census, 14	Computational			SF68 in sponge, placozoan,	Pan-metazoan

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
metazoan proteomes		Qualifies seed	Metazoan origin & breadth	cnidarians, mollusk, annelid, echinoderm, tick, Daphnia, 4 insects; only fly/ worm/yeast lost	
UniProt P78527	Database	Orientation	Family/ domain identity	Confirms PTHR11139:SF68; PIKK FAT/FATC/ PI3-4 kinase + DNA-PKcs N/CC domains	Human

PTHR11139 family census (provenance)

Organism	Lineage	PTHR11139 members	DNA-PKcs (SF68)
Homo sapiens	Vertebrata	6 subfamilies	**PRESENT** (P78527)
Anopheles gambiae	Insecta:Diptera	5	**PRESENT** (Q7Q293)
Aedes aegypti / *Tribolium* / *Apis* / *Bombyx*	Insecta	1 SF68 each	**PRESENT**
Amphimedon / *Trichoplax* / *Nematostella* / *Hydra*	Non-bilaterian	≥1	**PRESENT**
Crassostrea / *Capitella* / *Strongylocentrotus* / *Ixodes* / *Daphnia*	Protostome/ deuterostome	≥1	**PRESENT**
Drosophila melanogaster	Insecta:Diptera	8	**ABSENT** (0 SF68)
Caenorhabditis elegans	Nematoda	5	**ABSENT** (0 SF68)
Saccharomyces cerevisiae	Fungi	4	**ABSENT** (0 SF68)
Octopus bimaculoides	Mollusca	0 (incomplete proteome)	uninformative

GO Decision Table

GO aspect	Candidate term	Evidence basis	Recommended action (lead)
MF (core)	GO:0004677 DNA-dependent protein kinase activity	Extensive direct biochemistry (out of scope here)	**Retain** — unaffected by this hypothesis
BP (route)	regulation of macrophage proliferation / GO:0043277 apoptotic cell clearance	Gerlach 2021, mouse, context-specific	**Non-core**; annotate only with experimental evidence + cell-type qualifier, if at all
any	*(ortholog distribution)*	OrthoDB + UniProt/ PANTHER census (this work)	**No GO change** — evolutionary context only; at most a taxon-constraint note, never a function assertion
MF	protein binding	—	**Avoid** as a recommendation (uninformative)

GO Curation Implications

Lead (requires curator verification): treat `prkdc-animal-distribution` as non-core background context, not a GO-term change.

- The phylogenetic-distribution observation does **not** support adding, removing, generalizing, or specializing any **MF**/**BP**/**CC** term on human PRKDC. Ortholog loss in flies/worms is a property of those genomes, not evidence about the human gene product's function.
- The canonical, well-evidenced molecular function — *GO:0004677 DNA-dependent protein kinase activity* — is **retained** and unaffected by this hypothesis.
- If the efferocytosis route were annotated at all, defensible terms (e.g., *GO:0043277 apoptotic cell clearance* or a regulation-of-macrophage-proliferation BP) must carry **experimental evidence codes and a cell-type/species qualifier**, and must **not** be promoted to a core PRKDC function. The stimulus should be described as DNase2a-derived nucleotides, not intact DNA.

- **Do not** use "protein binding" as the outcome; it is uninformative and not what the evidence addresses.

Net: log the seed as orientation/mechanistic color; it is not a curation-actionable change to the PRKDC GO set.

Mechanistic Scope

- **Immediate molecular function tested:** DNA-PKcs serine/threonine kinase activity within a signaling cascade (→ mTORC2/Rictor → Myc). The direct gene-product activity is kinase signaling; the actual sensed species are **DNase2a-generated nucleotides**, placing DNA-PKcs downstream of DNA degradation rather than as a primary intact-DNA sensor.
- **Downstream phenotypes (not direct activity):** macrophage proliferation, apoptotic-cell clearance, tissue resolution, plaque stabilization — pathway/organismal outcomes partly inferred from loss-of-function (DNase2a/Rictor knockdown).
- **Evolutionary distribution:** a property of genomes, fully separable from the human gene product's biochemical function.

Evidence Base (Literature)

- *Efferocytosis induces macrophage proliferation to help resolve tissue injury.* **PMID: 34784501** (Gerlach et al., 2021). Establishes the efferocytosis-induced macrophage proliferation route and DNA-PKcs's role; clarifies the proximal stimulus is DNase2a-derived nucleotides. Verified snippet: *"nucleotides derived from the hydrolysis of apoptotic cell DNA by phagolysosomal DNase2a activate a DNA-PKcs-mTORC2/Rictor pathway that increases Myc to promote non-inflammatory macrophage proliferation."* **Supports Finding 2.**
- *Uncovering DNA-PKcs ancient phylogeny, unique sequence motifs and insights for human disease.* **PMID: 33035590** (Lees-Miller et al., 2021). Shows DNA-PKcs is ancient and broadly distributed, refuting vertebrate specificity. Verified snippet: *"the absence of DNA-PKcs from many model organisms has led to the assumption that DNA-PKcs is a vertebrate-specific PIKK. Here, we find that DNA-PKcs is widely distributed in invertebrates, fungi, plants, and protists."* **Supports Findings 1, 3, 4.**

Public bioinformatics resources used for direct presence/absence calls: OrthoDB (orthogroup `8724497at2759` / `5996771at33208`), UniProt (P78527 and per-organism proteomes), PANTHER family PTHR11139 / subfamily SF68.

Limitations and Knowledge Gaps

1. **Absence calls depend on proteome completeness.** True losses were confirmed only where the genome is complete and sister kinases are present (fly, worm, yeast). Incomplete proteomes (e.g., octopus, 0 family members) are uninformative, not evidence of loss. An exhaustive insect-wide loss map would require a per-proteome HMM sweep against assembled genomes.
2. **Sampling bias in OrthoDB.** The carrier set is bird-dominated; raw counts and clade-level absences must not be over-read. Iteration 3 demonstrated that beetle/bee/silkmoth OrthoDB absences were artifacts corrected by UniProt.
3. **No from-scratch phylogeny.** Conclusions rest on subfamily membership (PANTHER) and orthogroups (OrthoDB), not a maximum-likelihood tree with branch support; headline taxa are robust, but loss-event polarity would be strengthened by a formal reconstruction.
4. **Human-relevance of the efferocytosis route.** Gerlach 2021 is mouse/in-vitro; whether the DNase2a→DNA-PKcs→mTORC2→Myc axis operates identically in human macrophages, and merits a human-gene annotation, is unestablished.
5. **Mechanistic directness.** The proliferation phenotype relies substantially on knockdown of upstream (DNase2a) and parallel (Rictor) nodes; a catalytic-dead vs. kinase-active rescue would establish non-redundant DNA-PKcs requirement.

Proposed Follow-up Experiments / Actions

1. **Genome-level loss confirmation.** Run an HMMER/tBLASTn sweep with the PTHR11139:SF68 HMM against assembled *Drosophila* and *C. elegans* genomes (not just proteomes), and against "absent" taxa like octopus, to separate true loss from annotation gaps. Expected: no SF68 in fly/worm genomic DNA; SF68 recovered in octopus genome. *This is the single most informative confirmatory analysis.*

2. **Formal phylogeny + ancestral-state reconstruction.** Build a PTHR11139 maximum-likelihood tree highlighting SF68 to confirm metazoan-ancestral origin and polarity of the two independent loss events.
3. **Human efferocytosis validation.** Inhibit (NU7441) or knock out DNA-PKcs in primary human efferocytic macrophages; measure Myc induction and proliferation — distinguishes conserved-in-human from mouse-specific.
4. **Stimulus-specificity reconstitution.** Compare DNA-PKcs activation by DNase2a-digested nucleotides vs. intact dsDNA ends to confirm the nucleotide-sensing mechanism.
5. **Curation action.** Mark `prkdc-animal-distribution` as non-core/background; make no GO change on its basis. Candidate references to verify:

   P 34784501

   and

   P 33035590

   (snippets above). Pose the curator question: is ortholog distribution intended to *scope* a BP annotation (valid as a taxon-constraint note) or to *assert* a function (not valid)? Only the former is defensible.

Provenance

Computed via public OrthoDB REST (`data.orthodb.org/current`) and UniProt REST (`rest.uniprot.org`). Verified facts: PTHR11139:SF68 ↔ P78527; DNA-PKcs orthogroup present in 287 eukaryotic species (283 single-copy); *Drosophila melanogaster*, *C. elegans*, *S. cerevisiae* absent from the orthogroup; invertebrate carriers include *Branchiostoma lanceolatum*, *Acanthaster planci*, *Anopheles merus*, *Bradysia odoriphaga*. UniProt/PANTHER family census confirmed fly (8 members) and worm (5 members) retain TRRAP/mTOR/ATR/SMG1 but zero SF68 (true subfamily-specific loss); *Anopheles gambiae* retains SF68 (Q7Q293); phylum-broad census confirmed SF68 presence from sponges/cnidarians through most insects, with OrthoDB insect absences shown to be sampling artifacts.
